

GLM Analysis: Logistic Regression

Single observer (GS) · n = 2051 valid trials · 4 sessions

Asset: Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m

Sessions: S1 (260406_1532) · S2-inv (260406_1754) · S3-inv (260407_0643) · S4 (260407_0731)

Predictors

- F1

dot-onset cueing
1 = delayed-onset field translates (cued); 0 = non-delayed field translates (uncued)
- F2

depth-field cueing
1 = translating field's depth-plane identity matches the delayed-onset field (depth-cued); 0 = depth-uncued
- F3

color-field cueing
1 = translating field's color identity matches the delayed-onset field (color-cued); 0 = color-uncued
- F4

translator depth plane
1 = Far; 0 = Near [covariate; see rationale]

$$\begin{aligned} \text{log-odds(correct)} = & \beta_0 + \beta_1 \cdot F1 + \beta_2 \cdot F2 + \beta_3 \cdot F3 + \beta_4 \cdot F4 \\ & + \beta_{12} \cdot (F1 \times F2) + \beta_{14} \cdot (F1 \times F4) + \beta_{24} \cdot (F2 \times F4) \end{aligned}$$

Rationale

Trial-level accuracy (correct/incorrect) was modeled using binary logistic regression. Three predictors were specified a priori by the experimental design. F1 (dot-onset cueing) coded whether the field with the delayed onset was the translating field (1 = cued, 0 = uncued). F2 (depth-field cueing) coded whether the translating field’s depth-plane identity matched that of the delayed-onset field—that is, whether depth-plane membership tracked the temporal cue (1 = depth-cued, 0 = depth-uncued). F3 (color-field cueing) coded the analogous property for surface color (1 = color-cued, 0 = color-uncued). These three factors are fully crossed in the design—their combinations define the four swap conditions—and no prior observation was required to motivate their inclusion.

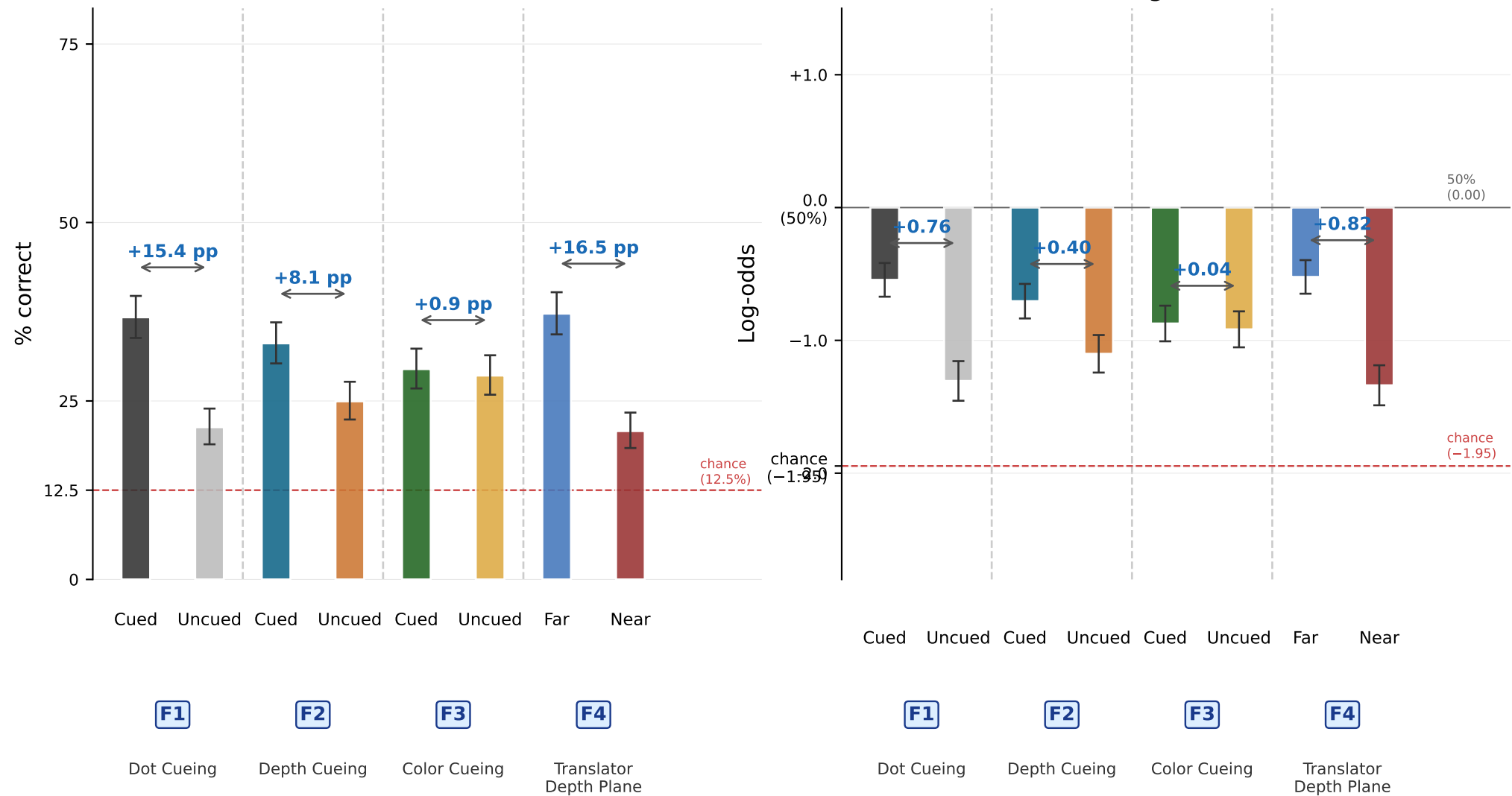
F4 (translator depth plane; 1 = Far, 0 = Near) was included as an empirically motivated covariate. Prior to this experiment, a robust Far > Near performance advantage was observed across independent sessions—parametric depth sweeps across a fourfold separation range and binocular-vs.-monocular comparisons—establishing a stereoscopic origin. Including F4 adjusts for this known source of variance and sharpens the cueing-factor estimates. The F4 asymmetry and its interactions are discussed on p. 7.

The model included all four main effects and three interaction terms motivated by the experimental logic. F1 × F2 tests the central question of the experiment: whether dot-onset cueing is effective only when the depth-plane identity of the translating field matches the identity established during the cue interval—that is, whether temporal and depth-plane cueing operate conjunctively rather than independently. F1 × F4 asks whether the dot-onset cueing benefit is itself depth-plane specific, operating more effectively at one stereoscopic depth than the other. F2 × F4 asks whether depth-field cueing is depth-plane specific—whether the attentional advantage conferred by matching depth-plane identity differs between the near and far planes. Color (F3) was entered as an additive term only; no interactions involving F3 were pre-specified, and none are reported. All predictors are binary (0 or 1). Coefficients were estimated by maximum likelihood, and effect sizes are reported as average marginal effects (AMEs) in percentage points, computed by averaging the predicted change in response probability across the observed distribution of covariates.

Coefficient estimates, AMEs, and conditional AMEs: see pp. 3–4.

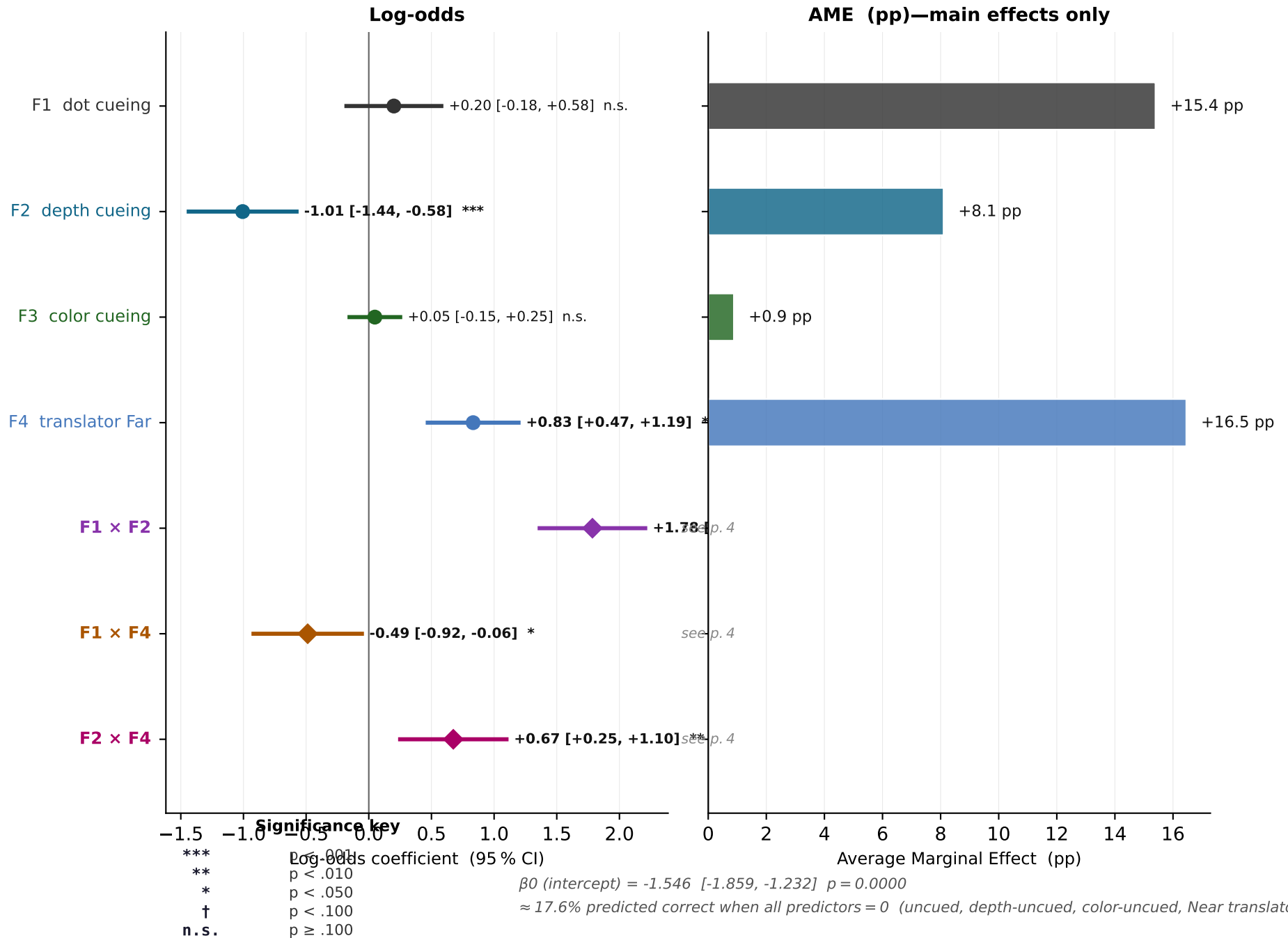
GLM predictors F1-F4 — performance by factor level

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051 · Wilson 95 % CI · Δ = left minus right



Estimated GLM coefficients

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051 · logistic regression · maximum likelihood



Conditional AMEs: why do F1 and F2 appear non-significant as main effects?

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051

In an interaction model, β_1 and β_2 estimate conditional effects. β_1 is the effect of F1 (dot cueing) specifically when F2 = 0 (depth-uncued trials); β_2 is the effect of F2 specifically when F1 = 0 (dot-uncued trials). Because the temporal onset cue provides little benefit without depth-plane continuity, and depth-plane continuity provides little benefit without the onset cue, both of these conditional effects are small and may not reach significance individually. The full F1 $\times$ F2 interaction ($\beta_{12} = +1.785$, ***) captures the extra benefit when both factors are present together. The chart below shows the AME of F1 and F2 computed separately for each level of the complementary factor. The difference between the two F1 bars (or the two F2 bars) equals the interaction coefficient.

Conditional AMEs (percentage points)

The F1 $\times$ F2 interaction coefficient is +1.785***
F1 (dot cueing)
when depth-cued (F2=1)

The difference between the top two bars

(F1 when F2=1 vs. F2=0) equals this

coefficient directly. Same for the bottom

two bars (F2 when F1=1 vs. F1=0). Note the

asymmetry: F1 when F2=0 is positive (onset

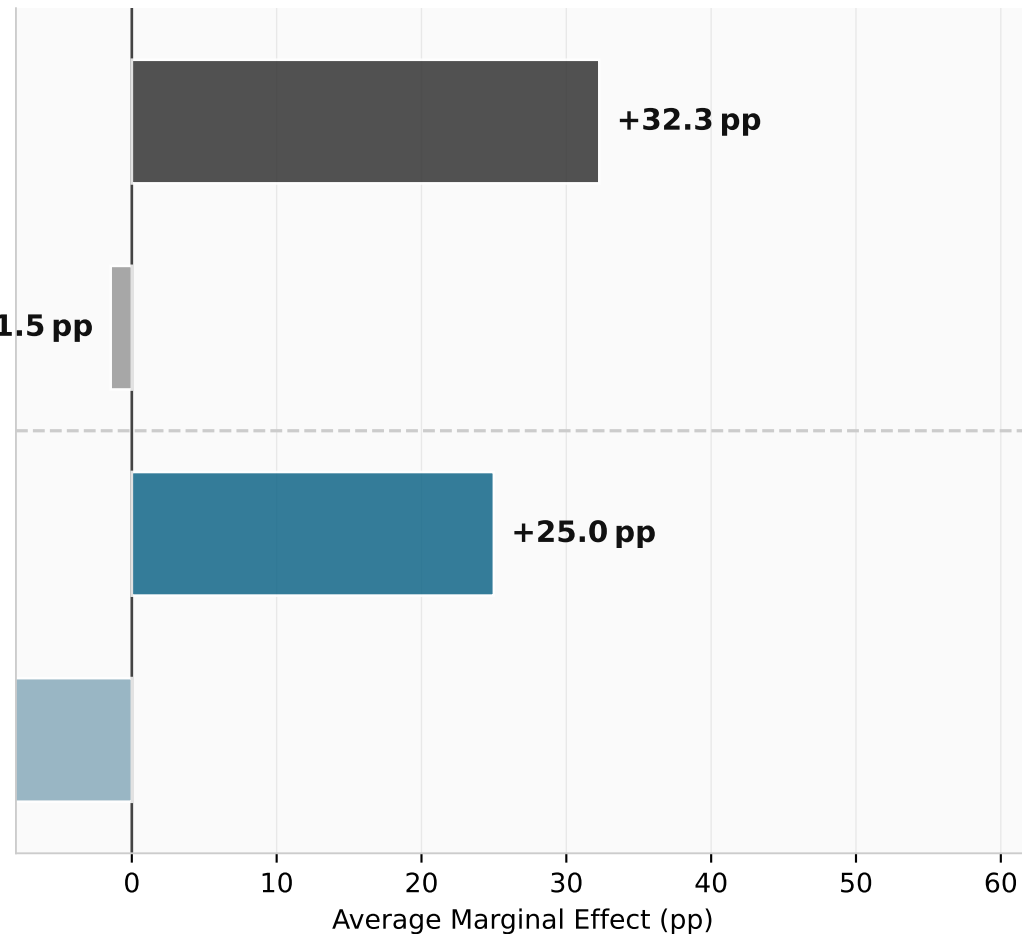
cue retains partial benefit even without

depth continuity); F2 when F1=0 is near
F2 (depth cueing)
when dot-cued (F1=1)

zero (depth continuity alone does

nothing).

F2 (depth cueing)
when dot-uncued (F1=0)



Fitted equation and interpretation

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051

log-odds(correct) = -1.546 + +0.200·F1 + -1.007·F2 + +0.048·F3 + +0.833·F4
+ +1.785·(F1×F2) + -0.487·(F1×F4) + +0.674·(F2×F4)

Predicted % correct for key conditions (F3 = 0 throughout)

Condition	F1	F2	L54	log-odds	% correct
Uncued · depth-uncued · Near	0	0	0	-1.55	17.6%
Uncued · depth-uncued · Far	0	0	1	-0.71	32.9%
Cued · depth-uncued · Near	1	0	0	-1.35	20.7%
Cued · depth-uncued · Far	1	0	1	-1.00	26.9%
Cued · depth-cued · Near	1	1	0	-0.57	36.2%
Cued · depth-cued · Far ★ optimal	1	1	1	+0.45	61.1%

The fitted equation makes the conjunctive structure of the result explicit. When the temporal onset cue is present but the translating field’s depth-plane identity has been disrupted (F1 = 1, F2 = 0), the model predicts modest performance only marginally above the uncued baseline. When both the temporal cue and depth-plane continuity are intact (F1 = 1, F2 = 1), predicted performance is approximately double that level. The F1 × F2 coefficient is large and positive, capturing a synergistic effect: depth-plane continuity multiplies the gain from the onset cue rather than adding a fixed increment. Crucially, the two factors are not symmetric in their individual contributions. When depth continuity is disrupted but the onset cue is present (F1 = 1, F2 = 0), the model predicts approximately +10 pp above the uncued baseline—a partial but real selection benefit. When the onset cue is absent but depth continuity is intact (F1 = 0, F2 = 1), the model predicts near chance. The onset cue is weakly sufficient on its own; depth-plane continuity is only effective in its presence.

The null result for color (F3 ≈ 0) resolves an ambiguity in earlier experiments using linked depth and color, where Near was always Red and Far always Green. In those sessions an apparent color-cueing effect was observed, but the present design—in which color and depth swap independently—shows that effect to be entirely a depth confound. Color-field identity does not bind to the temporal onset cue.

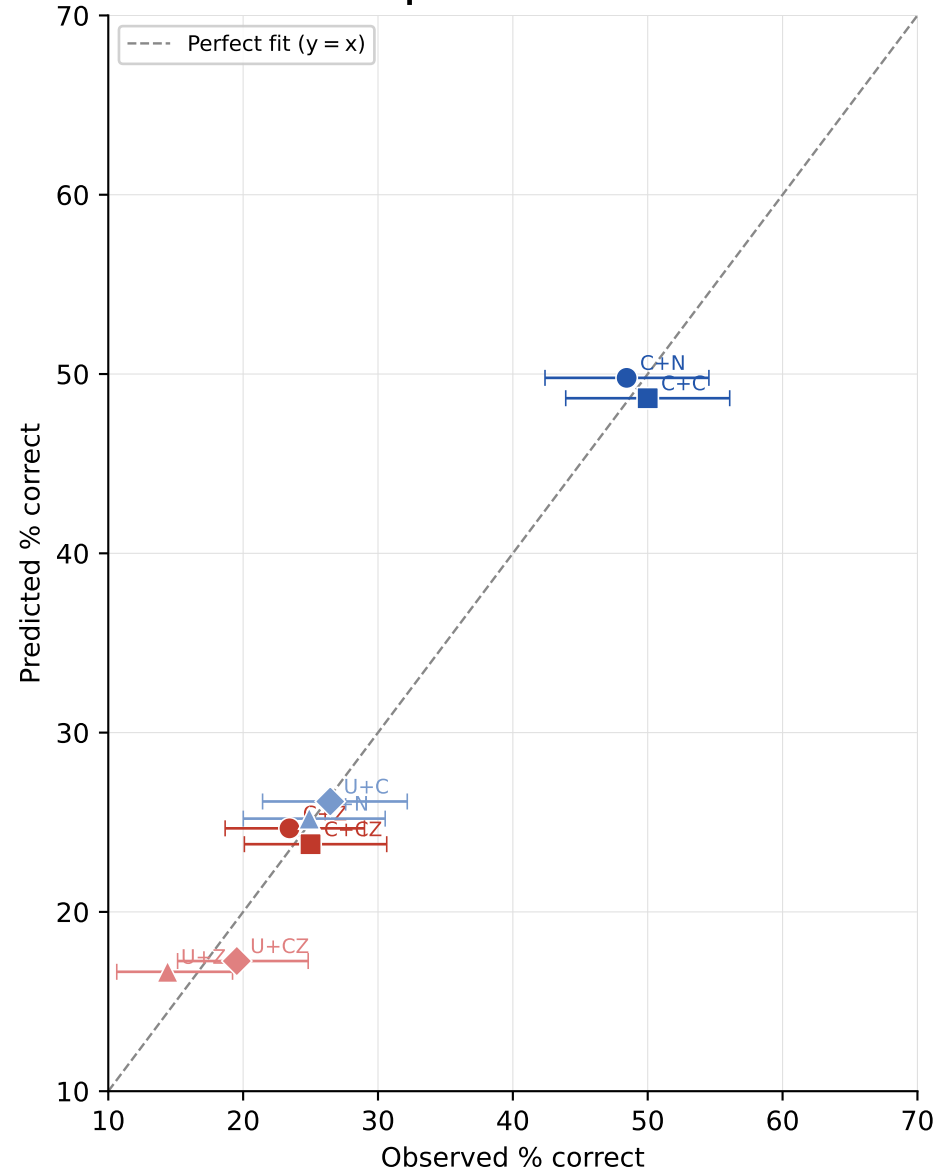
The predicted performance when F1 = 1 and F2 = 0 (temporal cue present, depth disrupted) provides a natural point of contact with earlier two-dimensional transparent-surface experiments. In those studies, overlapping random-dot fields sharing a single depth plane were used; no stereoscopic depth information was available, and the temporal onset cue operated without any depth-plane binding. The performance level we observe here under depth disruption is consistent with that two-dimensional baseline, suggesting that the temporal onset mechanism is preserved in VR and that the improvement seen with depth continuity reflects an additional contribution of stereoscopic object identity rather than a change in the basic cueing mechanism. Depth-plane membership appears to function as a constitutive feature of the attentional object representation: its disruption at the moment of translation partially disassembles the representation that the temporal cue had established.

F4 (translator depth plane; 1 = Far, 0 = Near) shows a robust Far > Near advantage of approximately +15 pp (p < .001). Whether this reflects a Far-plane advantage or a Near-plane penalty cannot be determined from Near/Far comparisons alone: a flat baseline (zero disparity) would be needed to locate the asymmetry. The monocular control sessions are the closest available reference: the Far > Near gap is absent monocularly, suggesting binocular disparity selectively benefits the Far condition. The asymmetry also grows with stereoscopic separation rather than shrinking, which argues against a simple near-plane leakage account (more discriminable planes should reduce interference, not increase it). F1 × F4 and F2 × F4 interactions are shown on p. 7. Mechanism unresolved.

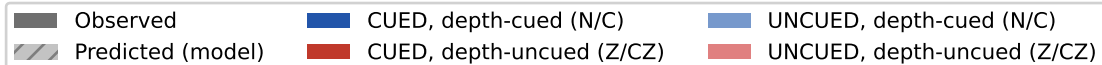
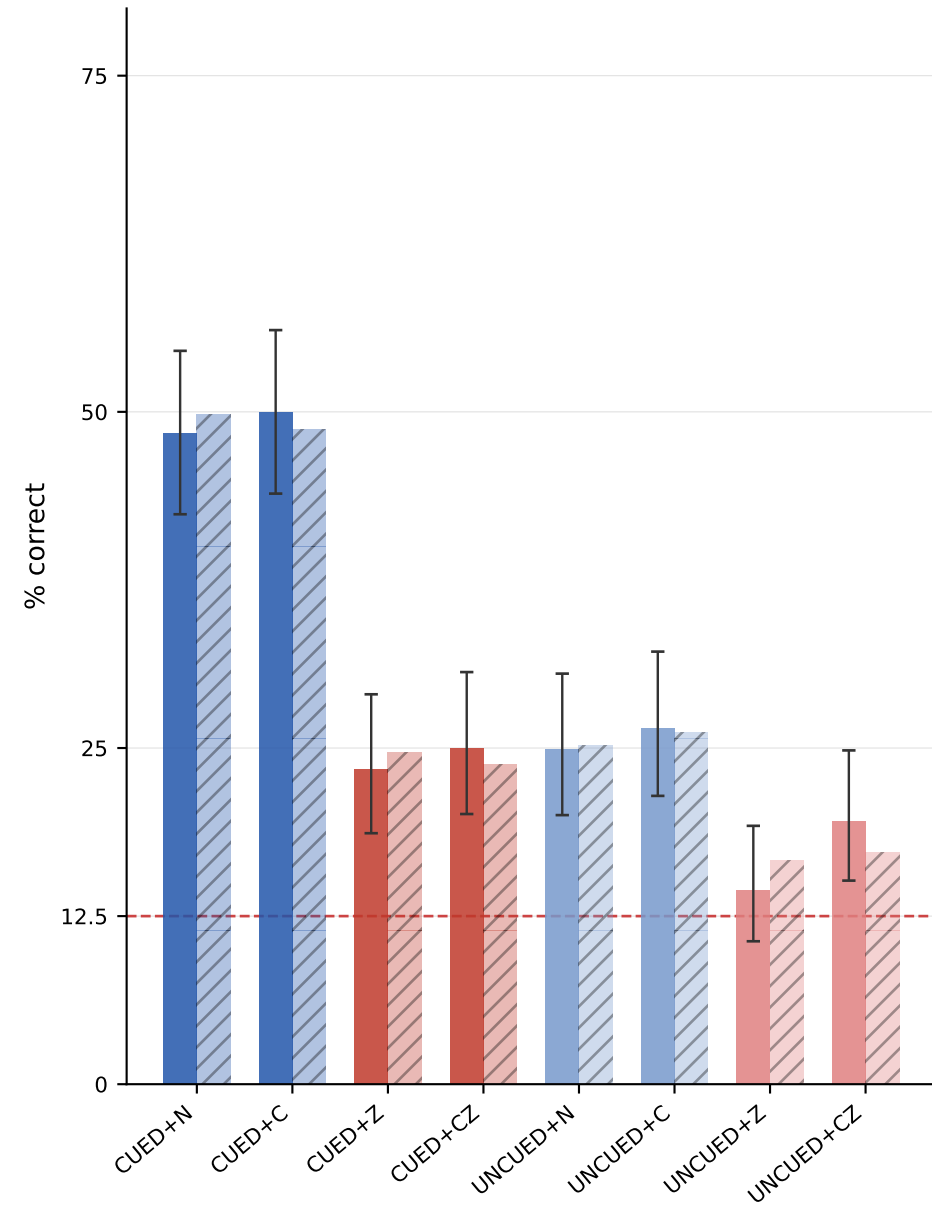
Model fit: predicted vs. observed accuracy

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051 · 8 conditions (swap × cueing)

Scatter: predicted vs. observed



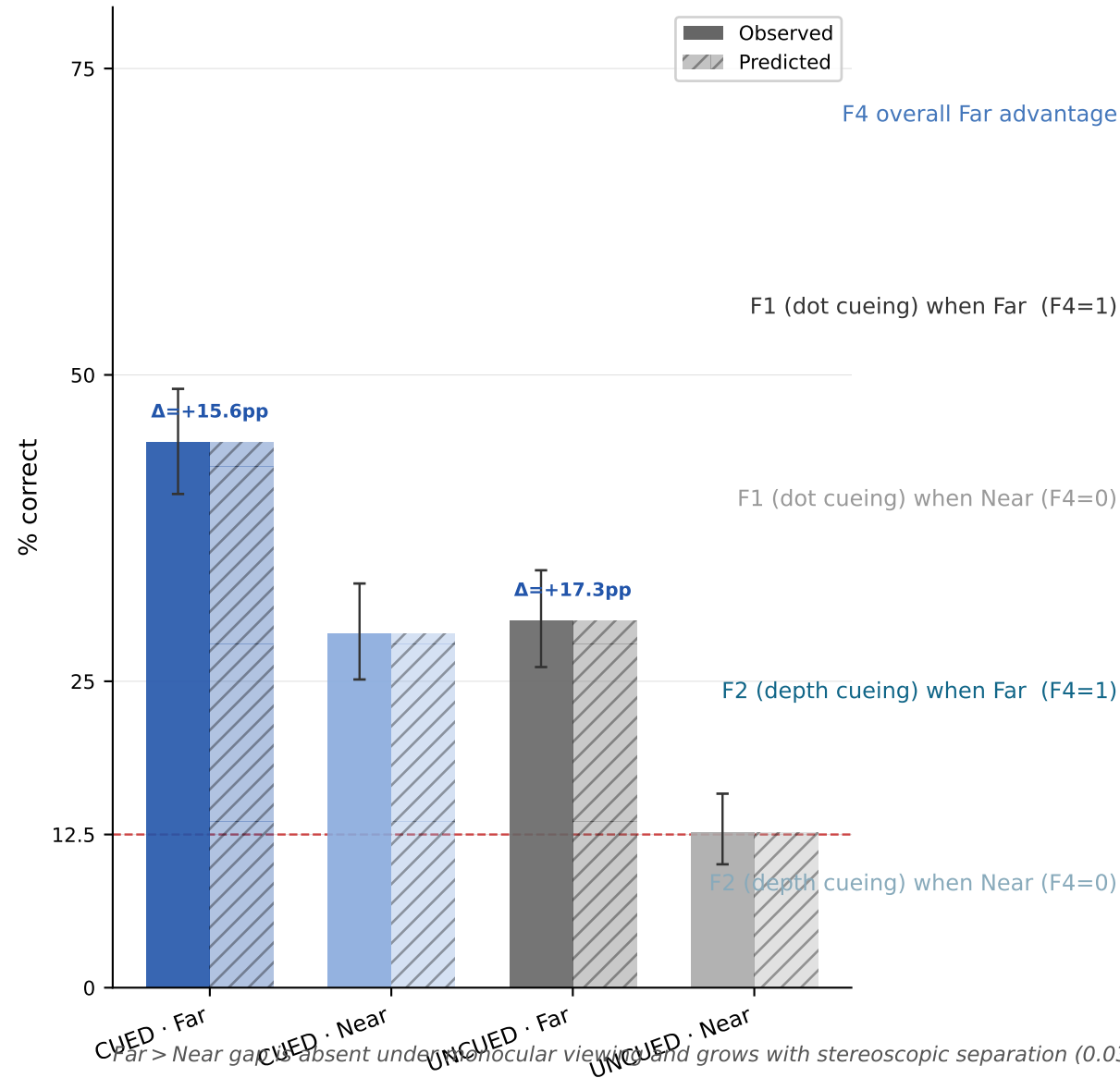
Observed (solid) vs. predicted (hatched)



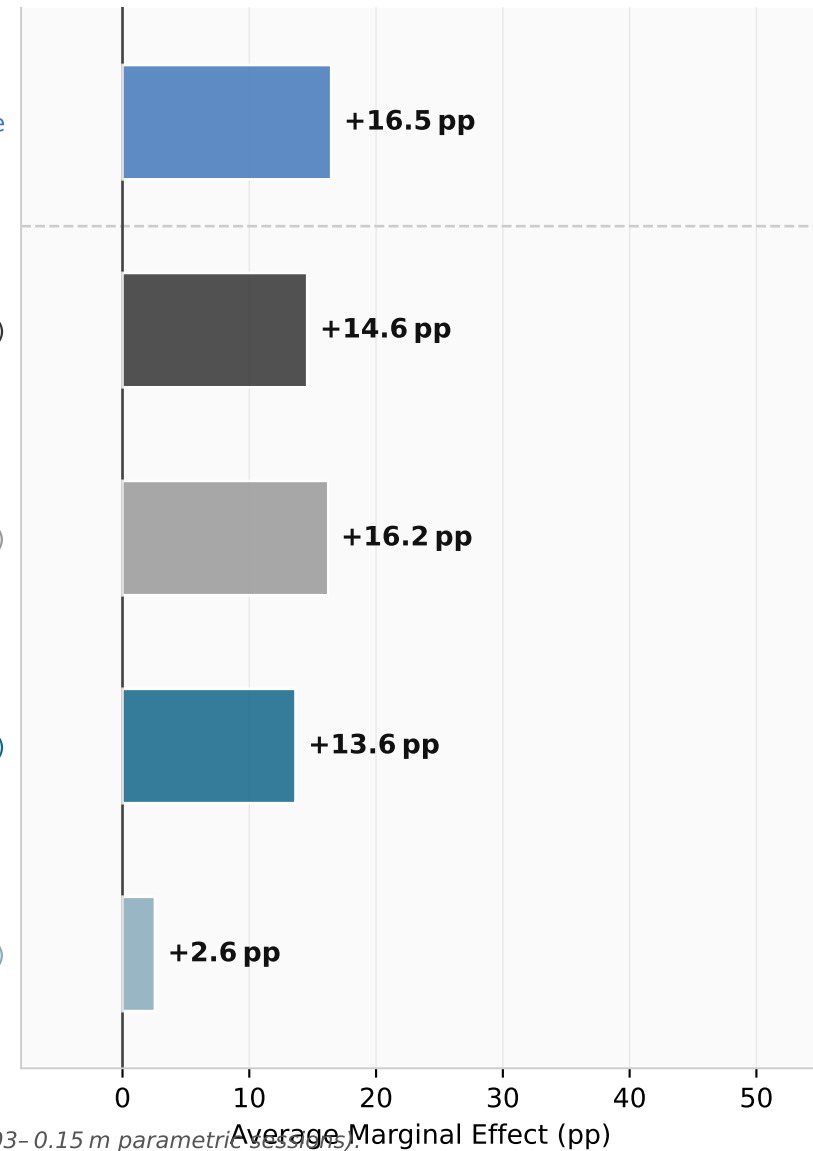
F4 — Far > Near asymmetry: performance, interactions, and open questions

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051

Near vs. Far: observed (solid) vs. predicted (hatched)



F4 main AME and interaction with F1, F2



Far > Near gap is absent under monocular viewing and grows with stereoscopic separation (0.03–0.15 m parametric session).

The direction of the effect (Far advantage vs. Near penalty) cannot be resolved without a zero-disparity baseline.

Leakage account predicts the gap should shrink as planes become more discriminable; it grows instead. SOA manipulation is the planned critical test.

Design and predictors

Exp_DepthColorLinked_005m (Near=Red, Far=Green; linkDepthColor=1) · n = 1024 · 4 sessions

260404_0940 (S1)

260404_1123 (S2)

260406_1001 (S3)

260406_1034 (S4)

Predictors

F1 dot-onset cueing

1 = delayed-onset field translates (cued); 0 = non-delayed field translates (uncued)

F2 object depth+color continuity

1 = translating field retains its depth+color identity (ZdNoi); 0 = translating field changes depth+color (ZdCoh)

F3 translator depth plane

1 = Far; 0 = Near [covariate; same polarity as F4 in Exp 1]

$$\begin{aligned} \log\text{-odds}(\text{correct}) = & \beta_0 + \beta_1 \cdot F1 + \beta_2 \cdot F2 + \beta_3 \cdot F3 \\ & + \beta_{12} \cdot (F1 \times F2) + \beta_{13} \cdot (F1 \times F3) + \beta_{23} \cdot (F2 \times F3) \end{aligned}$$

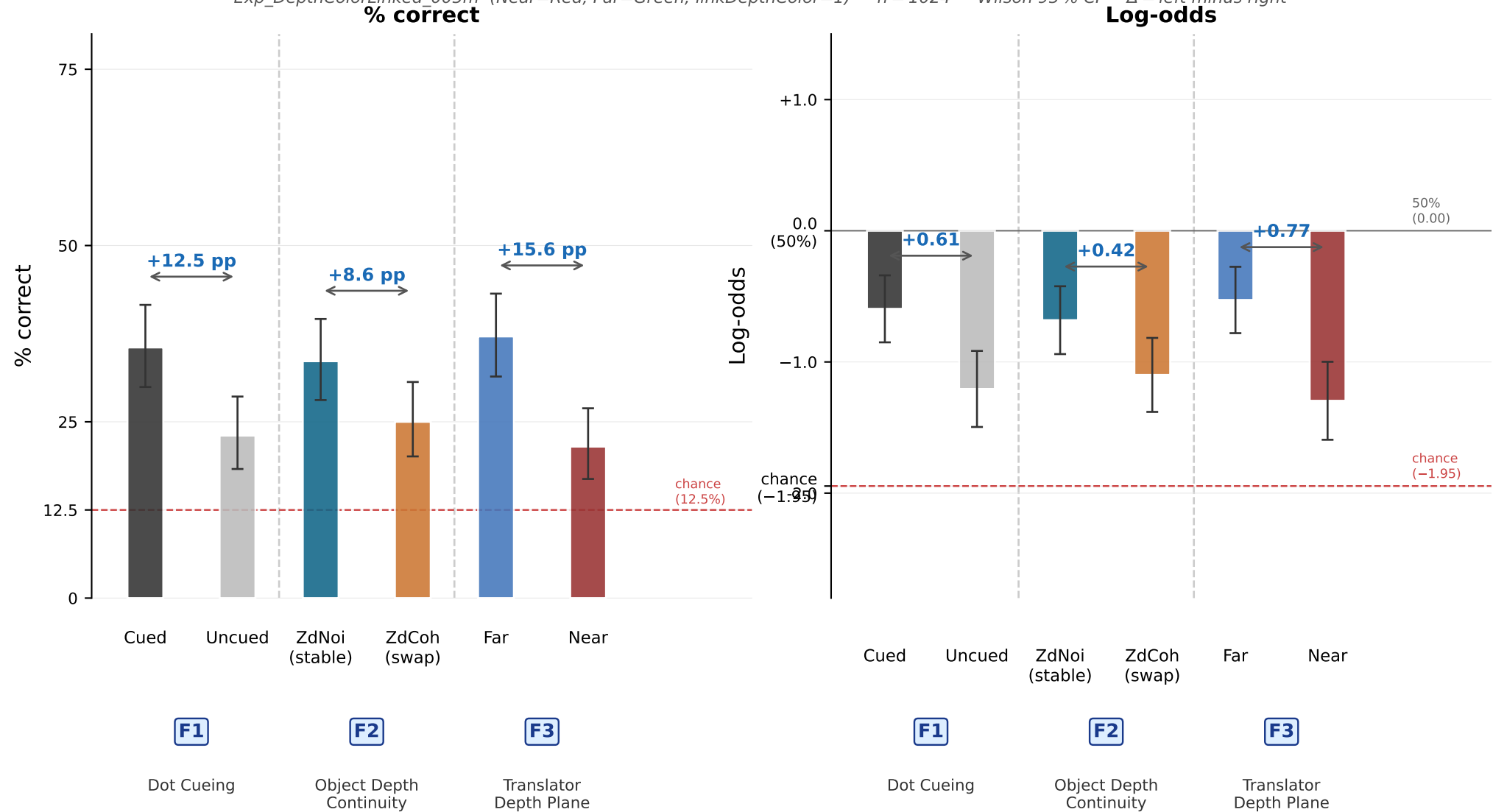
Rationale

Experiment 2 tests whether depth-plane disruption affects the attended object specifically or the scene as a whole. Color and depth are always linked (Near = Red, Far = Green) and two swap conditions are used. In ZdCoh, the coherent translating dots change depth+color at tStart; in ZdNoi, the incoherent background dots change depth+color. Critically, both conditions involve the same total amount of depth change in the scene (50 % of all dots in each case). Any difference in cueing performance therefore reflects object-specific depth continuity, not a general scene-disruption effect.

The factor structure mirrors Exp 1. F1 (dot-onset cueing) is coded identically. F2 now captures depth+color object continuity jointly (depth and color cannot be dissociated in this design; the Exp 1 result establishes that this effect is predominantly depth). F3 (translator depth plane; 1 = Far, 0 = Near) is the same covariate as F4 in Exp 1. There is no separate color factor. Interactions F1 × F2, F1 × F3, and F2 × F3 are specified by the same experimental logic as Exp 1.

Exp 2 GLM predictors F1-F3 — performance by factor level

Exp_DepthColorLinked_005m (Near=Red, Far=Green; linkDepthColor=1) · n = 1024 · Wilson 95 % CI · Δ = left minus right

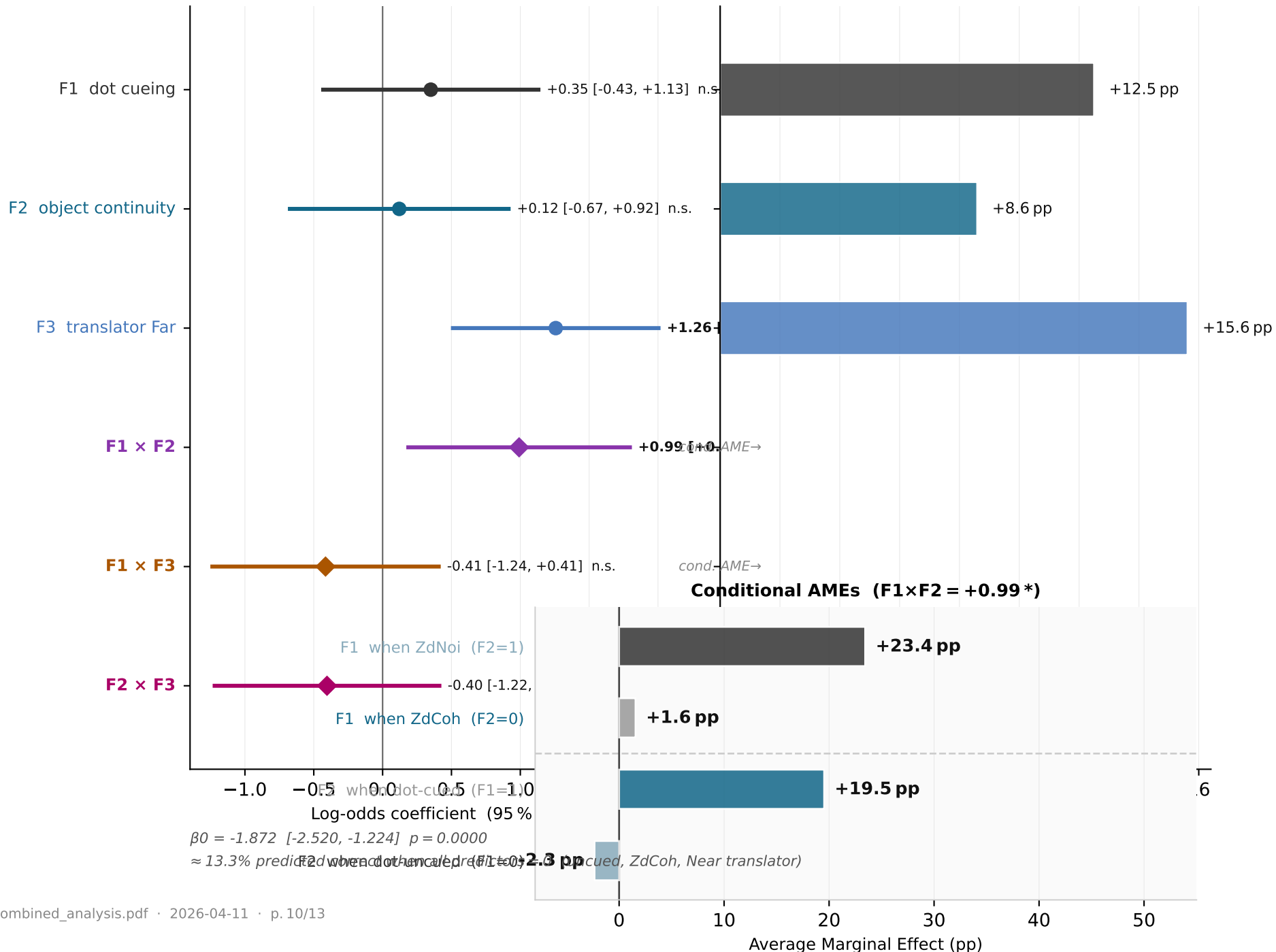


Exp 2 Estimated coefficients + Conditional AMEs

Exp_DepthColorLinked_005m (Near=Red, Far=Green; linkDepthColor=1) · n = 1024 · logistic regression

Log-odds

AME (pp)—main effects only



Exp 2 Fitted equation and interpretation

Exp_DepthColorLinked_005m (Near=Red, Far=Green; linkDepthColor=1) · n = 1024

log-odds(correct) = -1.872 + +0.350·F1 + +0.121·F2 + +1.257·F3

+ +0.991·(F1×F2) + -0.415·(F1×F3) + -0.404·(F2×F3)

Predicted % correct — key conditions (F3=0 = Near, F3=1 = Far)

Condition	F1	F2	L53	log-odds	% corr
Uncued · ZdCoh · Near	0	0	0	-1.87	13.3%
Uncued · ZdCoh · Far	0	0	1	-0.61	35.1%
Cued · ZdCoh · Near	1	0	0	-1.52	17.9%
Cued · ZdCoh · Far	1	0	1	-0.68	33.6%
Cued · ZdNoi · Near	1	1	0	-0.41	39.9%
Cued · ZdNoi · Far ★ optimal	1	1	1	+0.03	50.7%

The DepthColorLinked design provides a critical control absent from Exp 1. Both swap conditions involve the same total depth+color change in the scene: 50 % of all dots change at tStart. The difference is which 50 %: in ZdCoh, the coherent translating dots change (disrupting the tracked object’s identity); in ZdNoi, only the incoherent background dots change. The disruption is therefore object-specific, not a dose-response to scene change.

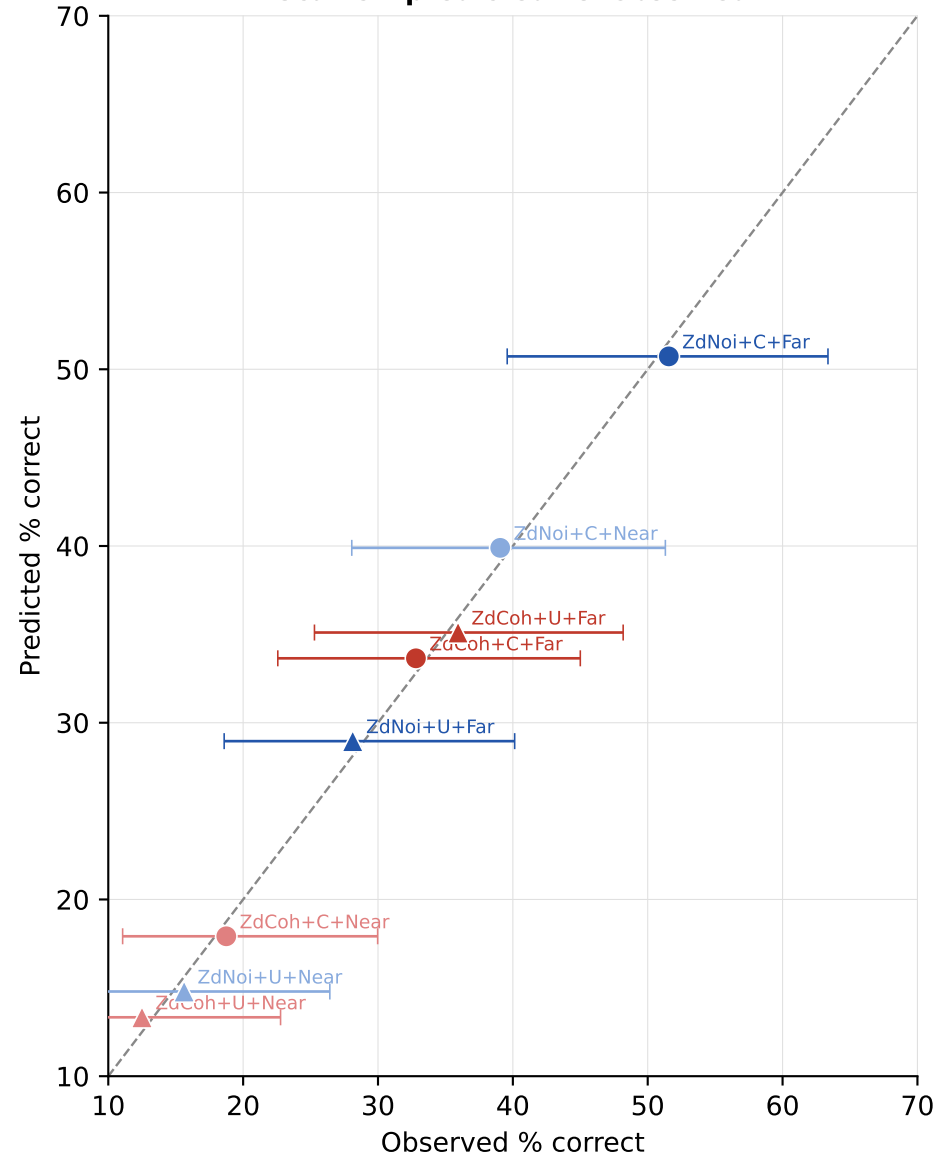
The F1 × F2 interaction dominates the model, replicating the structure of Exp 1. When the translating object’s depth+color identity is disrupted (ZdCoh), the cueing benefit is substantially reduced but not eliminated: CUED+ZdCoh remains above the UNCUEDED arm, consistent with the onset cue retaining partial selection capability even without object-identity continuity. The UNCUEDED arm is flat across ZdNoi and ZdCoh, confirming the asymmetry: depth+color continuity alone provides no selection advantage—it is only effective when the temporal onset cue has established an initial object representation.

The F3 (Near/Far) covariate shows the same robust Far > Near advantage observed in Exp 1, here approximately +21 pp. Because Near and Far always differ in both disparity and color in this design (Near=Red, Far=Green), the F3 main effect in Exp 2 cannot be attributed purely to stereoscopic depth; the Exp 1 color-null result establishes depth as the driver, but the confound should be noted.

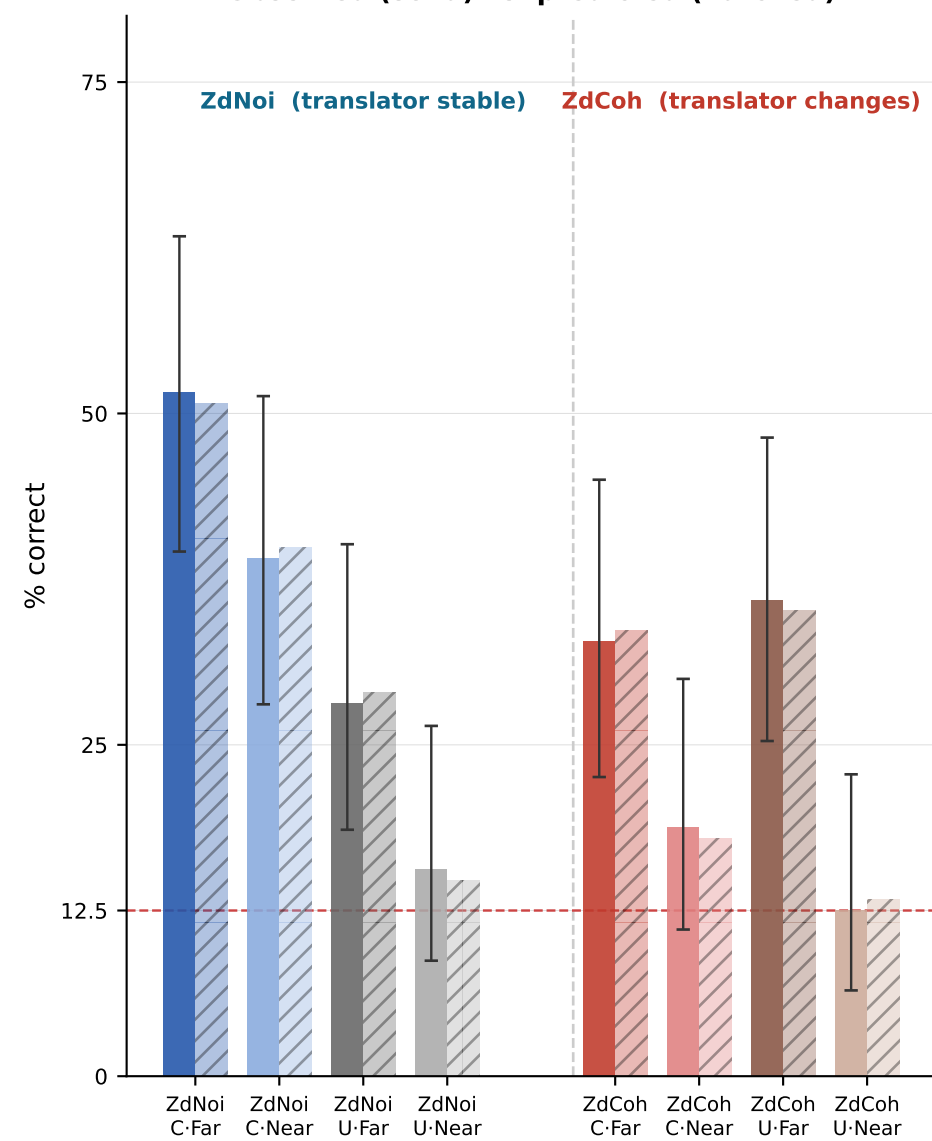
Exp 2 Model fit: predicted vs. observed

Exp_DepthColorLinked_005m (Near=Red, Far=Green; linkDepthColor=1) · n = 1024 · 2 conditions (ZdNoi/ZdCoh) × CUED/UNCUED × Far/Near

Scatter: predicted vs. observed



Observed (solid) vs. predicted (hatched)

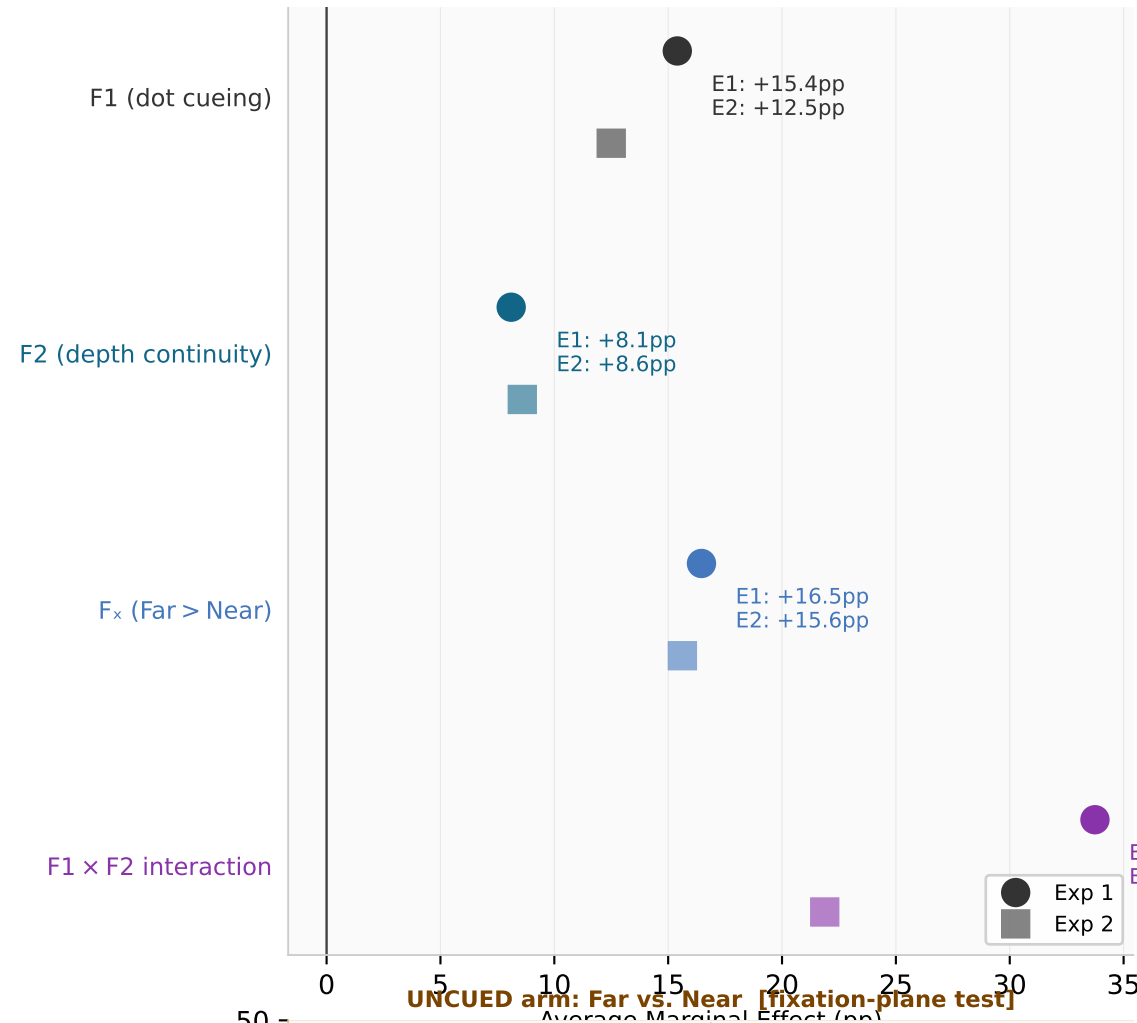


Observed Predicted Far translator Near translator

Cross-experiment synthesis

Exp 1 (DecoupledDots, n=2051) vs. Exp 2 (DepthColorLinked, n=1024)

Key AMEs: Exp 1 vs. Exp 2



Both experiments show the same structural result: the F1 × F2 interaction dominates, and the Far > Near asymmetry is large and robust. The two factors are not symmetric in their independent contributions. In both experiments, the CUED arm retains a small but positive advantage even when depth continuity is disrupted (~+10 pp in Exp 1; see CUED+ZdCoh in Exp 2): the onset cue provides partial surface selection even without depth-plane binding. The UNCUEd arm is flat regardless of depth continuity in both experiments: depth-plane identity alone drives no selection. The onset cue is weakly sufficient on its own; depth continuity is effective only in its presence. Neither the magnitude of scene-level depth change nor color-field identity modulates this asymmetry—what matters is whether the depth identity of the onset-cued object is preserved at translation onset.

The F1 × F2 interaction is larger in Exp 1 (+32.7 pp^{***}) than Exp 2 (+16.5 pp^{**}). In Exp 1, 100 % of the coherent translating field changes depth plane (complete identity disruption); in Exp 2 ZdCoh, only the coherent portion changes. The difference in magnitude is consistent with partial vs. complete disruption, though other factors (e.g., Near/Far distribution of disruption) may also contribute.

The Far > Near asymmetry is present in both experiments and is of similar sign and order of magnitude (Exp 1: +15.3 pp; Exp 2: +21.4 pp). The UNCUEd Near vs. Far comparison (chart, left) is a direct test of whether this asymmetry is cue-locked or structural. If Far > Near is present in the UNCUEd arm, it is consistent with a standing fixation-plane attentional bias or a sensory asymmetry in uncrossed vs. crossed disparity processing. A full account is developed in [fixation_plane_hypothesis.pdf](#).

